Exercise - 14.4

Integrate

1. $\int \frac{9 \cdot dx}{(2x+1)(x+5)}$

Solution:

Let $\frac{9}{(2x+1)(x+5)} = \frac{A}{(2x+1)} + \frac{B}{(x+5)}$

or $9 = A(x+5) + B(2x+1)$

or $9 = (A+2B)x + (5A+B)$

Equating the coefficient of x and constant terms we get.

$A + 2B = 0$ — (i)

$5A + B = 9$ — (ii)

$\therefore A = 2$ and $B = -1$

$\therefore \frac{9}{(2x+1)(x+5)} = \frac{2}{(2x+1)} + \frac{(-1)}{(x+5)}$

Now, Integrating both sides:

$$\int \frac{9 \cdot dx}{(2x+1)(x+5)} = \int \frac{2 \cdot dx}{2x+1} - \int \frac{1 \cdot dx}{(x+5)}$$

$$= \text{Log}(2x+1) - \text{Log}(x+5) + C$$

2. $\int \frac{13 \cdot dx}{(3x+4)(4x+1)}$

Solution:

Let $\frac{13}{(3x+4)(4x+1)} = \frac{A}{(3x+4)} + \frac{B}{(4x+1)}$

or, $9 = A(4x+1) + B(3x+4)$

Equating the coefficient of x and constant terms;

$4A + 3B = 0$, $(A+4B) = 13$

on solving them we get, $A = -3$ and $B = 4$

$$\therefore \frac{13}{(3x+4)(4x+1)} = \frac{-3}{(3x+4)} + \frac{4}{(4x+1)}$$

Integrating both sides;

$$\int \frac{13}{(3x+4)(4x+1)}\,dx = \int \frac{-3\,dx}{(3x+4)} + 4\int \frac{dx}{4x+1}$$

$$= -\log(3x+4) + \log(4x+1)$$

$$= \log(4x+1) - \log(3x+4) + c$$

3. $$\int \frac{1}{(x+2)(x+3)^2}\,dx$$

Solution:

Let $$\frac{1}{(x+2)(x+3)^2} = \frac{A}{(x+2)} + \frac{B}{(x+3)} + \frac{C}{(x+3)^2}$$

or $$1 = A(x+3)^2 + B(x+2)(x+3) + C(x+2)$$

Equating the component of x and the constant terms;

We get $A = 1$, $B = -1$, $C = -1$

$$\therefore \frac{1}{(x+2)(x+3)^2} = \frac{1}{(x+2)} - \frac{1}{(x+3)} - \frac{1}{(x+3)^2}$$

Integrating both sides;

$$\int \frac{1}{(x+2)(x+3)^2}\,dx = \int \frac{1\,dx}{(x+2)} - \int \frac{dx}{(x+3)} - \int \frac{dx}{(x+3)^2}$$

$$= \log(x+2) - \log(x+3) - \frac{(x+3)^{-1}}{-1}$$

$$= \log(x+2) - \log(x+3) + \frac{1}{(x+3)} + c$$

$$= \log\left(\frac{x+2}{x+3}\right) + \frac{1}{(x+3)} + c$$

4. $\int \frac{3x \cdot dx}{(x-a)(x-b)}$

Solution:

Let $\frac{x}{(x-a)(x-b)} = \frac{x-a+a}{(x-a)(x-b)}$

or, ~~$3x = A(x-b) + B(x-a)$~~

$= \frac{(x-a)}{(x-a)(x-b)} + \frac{a}{(x-b)(x-a)}$

Then, $3\int \frac{x \cdot dx}{(x-a)(x-b)} = 3\left[\int \frac{1\,dx}{(x-b)} + \int \frac{a\,dx}{(x-a)(x-b)}\right]$

$= 3\left[\log(x-b) + \frac{a}{a-b}\int \frac{(a-b)\,dx}{(x-a)(x-b)}\right] + C$

$= 3\left[\log(x-b) + \frac{a}{a-b}\left\{\int \frac{(x-b)-(x-a)}{(x-a)(x-b)}\right\}\right]$

$= 3\left[\log(x-b) + \frac{a}{(a-b)}\left\{\log(x-a) - \log(x-b)\right\}\right]$

$= 3\left[\log(x-b) + \frac{a}{a-b}\log\left(\frac{x-a}{x-b}\right)\right] + C$

5. $\int \frac{(x+1)\,dx}{x^2-2x-8}$

Solution:

$= \int \frac{(x+1)\,dx}{(x-4)(x+2)}$

Let, $\frac{x+1}{(x-4)(x+2)} = \frac{A}{(x-4)} + \frac{B}{(x+2)}$

or, $x+1 = A(x+2) + B(x-4)$

or, $x+1 = Ax + 2A + Bx - 4B$

or, $x+1 = (A+B)x + (2A-4B)$

On comparing, we get:

$A+B = 1$ — (i)

$2A-4B = 1$ — (ii)

On Solving (i) and (ii) we get;

$$A = \frac{5}{6} \quad \text{and} \quad B = \frac{1}{6}$$

Hence, $$\frac{x+1}{(x-4)(x+2)} = \frac{5}{6(x-4)} + \frac{1}{6(x+2)}$$

Integrating both sides:

$$\int \frac{x+1}{(x-4)(x+2)}\,dx = \frac{5}{6}\int \frac{dx}{(x-4)} + \frac{1}{6}\int \frac{dx}{(x+2)}$$

$$= \frac{5}{6}\log(x-4) + \frac{1}{6}\log(x+2) + C$$

$$= \frac{1}{6}\left[5\log(x-4) + \log(x+2)\right] + C$$

6. $$\int \frac{5\,.\,dx}{(x+5)(2x^2+5)}$$

Solution:

Let $$\frac{5}{(x+5)(2x^2+5)} = \frac{A}{(x+5)} + \frac{Bx+C}{(2x^2+5)}$$

$$\text{or, } 5 = A(2x^2+5) + (Bx+C)(x+5)$$

$$\text{or, } 5 = 2Ax^2 + 5A + Bx^2 + 5Bx + Cx + 5C$$

$$\text{or, } 5 = 2Ax^2 + Bx^2 + 5Bx + Cx + 5A + 5C$$

$$\text{or, } 5 = (2A+B)x^2 + (5B+C)x + (5A+5C)$$

On Comparing the like terms, we get;

$$2A + B = 0 \quad \text{—(i)}$$

$$5B + C = 0 \quad \text{—(ii)}$$

$$5A + 5C = 5 \quad \text{—(iii)}$$

On Solving them we get;

$$A = \frac{1}{11}, \quad B = \frac{-2}{11}, \quad C = \frac{10}{11}$$

Then,

$$\frac{5}{(x+5)(2x^2+5)} = \frac{1}{11(x+5)} - \frac{(2x+10)}{11(2x^2+5)}$$

Date: / /

Integrating both side, we get;

$$\int \frac{5\,dx}{(x+5)(2x^2+5)} = \frac{1}{11}\int \frac{dx}{(x+5)} - \frac{1}{11}\int \frac{dx\,(2x+10)}{(2x^2+5)}$$

$$= \frac{1}{11}\int \frac{dx}{(x+5)} - \left[\int \frac{2x\,dx}{2x^2+5} - \int \frac{10\,dx}{2x^2+5}\right]\frac{1}{11}$$

$$= \frac{1}{11}\log(x+5) - \frac{1}{11}\left[\frac{\log(2x^2+5)}{2} - \int \frac{10\,dx}{2x^2+5}\right]$$

$$= \frac{1}{11}\log(x+5) - \frac{1}{22}\log(2x^2+5) + \frac{5}{11}\int \frac{dx}{x^2+(\sqrt{5/2})}$$

$$= \frac{1}{11}\log(x+5) - \frac{1}{22}\log(2x^2+5) + \frac{5.1}{11\sqrt{5/2}}\cdot \tan^{-1}\frac{x}{\sqrt{5/2}}$$

$$= \frac{1}{11}\log(x+5) - \frac{1}{22}\log(2x^2+5) + \frac{\sqrt{2}\cdot 5}{11\sqrt{5}}\cdot \tan^{-1}\sqrt{\frac{2}{5}}x$$

$$= \frac{1}{11}\left[\log(x+5) - \frac{1}{2}\log(2x^2+5) + \sqrt{10}\tan^{-1}\sqrt{\frac{2}{5}}x\right] + C$$

7. $\int \frac{1}{(a^2+x^2)(b^2+x^2)}\,dx$

Solution:

Let $\frac{1}{(a^2+x^2)(b^2+x^2)} = \frac{Ax+B}{(a^2+x^2)} + \frac{Cx+D}{(b^2+x^2)}$

or, $1 = (Ax+B)(b^2+x^2) + (Cx+D)(a^2+x^2)$

or, $1 = Axb^2 + Ax^3 + Bb^2 + Bx^2 + Cxa^2 + Cx^3 + Da^2 + Dx^2$

or, $1 = Ax^3 + Cx^3 + Bx^2 + Dx^2 + Axb^2 + Cxa^2 + Bb^2 + Da^2$

or, $1 = (A+C)x^3 + (B+D)x^2 + (Ab^2 + Ca^2)x + Bb^2 + Da^2$

Comparing the like terms, we get;

$A + C = 0$

or, $A = -C$

$\therefore A = 0$

$B + D = 0$

or, $B = -D$

or, $B = \frac{-1}{a^2 - b^2}$

$Ab^2 + Ca^2 = 0$

or, $-Cb^2 + Ca^2 = 0$

or, $C(a^2 - b^2) = 0$

$\therefore C = 0$

$Bb^2 + Da^2 = 1$

or, $-Db^2 + Da^2 = 1$

or, $D(a^2 - b^2) = 1$

$\therefore D = \frac{1}{a^2 - b^2}$

Then,

$$\frac{1}{(a^2+x^2)(b^2+x^2)} = \frac{-1}{(a^2-b^2)(a^2+x^2)} + \frac{1}{(a^2-b^2)(b^2+x^2)}$$

Integrating both sides we get;

$$\int \frac{1 \cdot dx}{(a^2+x^2)(b^2+x^2)} = \frac{1}{a^2-b^2}\left[\int \frac{dx}{x^2+b^2} - \int \frac{dx}{a^2+x^2}\right]$$

$$= \frac{1}{a^2-b^2}\left[\frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right) - \frac{1}{a}\tan^{-1}\frac{x}{a}\right] + C$$

8. $\int \frac{1-x}{x^2+x^3}\,dx$

Solution:

Let $\frac{1-x}{x^2(1+x)} = \frac{Ax+B}{x^2} + \frac{C}{1+x}$

or, $1 - x = (Ax + B)(1 + x) + C(x^2)$

or, $1 - x = Ax + Ax^2 + Cx^2 + B + Bx$

or, $1 - x = Ax^2 + Cx^2 + Bx + Ax + B$

or, $1 - x = (A + C)x^2 + (A + B)x + B$

On Comparing like terms, we get

$A + C = 0$ —— (i)

$A + B = -1$ —— (ii)

$B = 1$ —— (iii)

On solving them we get;

$A = -2$, $B = 1$, $C = 2$

Then,

$$\frac{1-x}{x^2(1+x)} = \frac{-2x+1}{x^2} + \frac{2}{1+x}$$

Integrating both sides;

$$\int \frac{1-x}{x^2+x^3} = \int \frac{(-2x+1)\,dx}{x^2} + 2\int \frac{dx}{1+x}$$

$$= -\int \frac{(2x-1)\,dx}{x^2} + 2\log(1+x) + C$$

$$= -\int \frac{2x\,dx}{x^2} + 1\int \frac{dx}{x^2} + 2\log(1+x) + C$$

$$= -2\log x - x^{-1} + 2\log(1+x) + C$$

$$= 2\log(1+x) - 2\log x - \frac{1}{x} + C$$

$$= 2\log\left(\frac{1+x}{x}\right) - \frac{1}{x} + C$$

9. $\int \frac{(x^2-1)\,dx}{x^4+x^2+1}$

Solution:

$$= \int \frac{\cancel{\text{let}}\ x^2-1}{(x^4+x^2+1)}$$

$$= \int \frac{x^2-1}{(x^2+x+1)(x^2-x+1)}$$

Let, $\frac{x^2-1}{(x^2+x+1)(x^2-x+1)} = \frac{Ax+B}{(x^2+x+1)} + \frac{Cx+D}{(x^2-x+1)}$

or, $x^2-1 = (Ax+B)(x^2-x+1) + (Cx+D)(x^2+x+1)$

or, $x^2-1 = Ax^3 - Ax^2 + Ax + Bx^2 - Bx + B + Cx^3 + Cx^2 + Cx + Dx^2 + Dx + D$

or, $x^2-1 = Ax^3 + Cx^3 + Cx^2 - Ax^2 + Bx^2 + Ax - Bx + Cx + Dx + Dx^2 + D$

or, $x^2-1 = (A+C)x^3 + (B-A+C+D)x^2 + (A+C+D-B)x + (B+D)$

On comparing like terms, we get;

$A+C=0$ —— (I)

$B-A+C+D=1$ —— (II)

$B+D=-1$ —— (III)

$A+C+D-B=0$ —— (IV)

On solving the above equation we get;

$A=-1$, $B=\frac{-1}{2}$, $C=1$ and $D=\frac{-1}{2}$

Then,

$$\int \frac{x^2-1}{(x^2+x+1)(x^2-x+1)}\,dx = \int \frac{\left(-x-\frac{1}{2}\right)}{(x^2+x+1)} + \frac{\left(x-\frac{1}{2}\right)}{(x^2-x+1)}\,dx$$

$$= \int \frac{(-2x-1)\,dx}{2(x^2+x+1)} + \int \frac{(2x-1)\,dx}{2(x^2-x+1)}$$

$$= \frac{1}{2}\left[-\int \frac{(2x+1)\,dx}{(x^2+x+1)} + \int \frac{(2x-1)\,dx}{x^2-x+1}\right]$$

$$= \frac{1}{2}\left[-\int \frac{dy}{y} + \int \frac{dz}{dz}\right]$$

$$= \frac{1}{2}\left[\log z - \log y\right] + C$$

$$= \frac{1}{2}\log\frac{z}{y} + C$$

$$= \frac{1}{2}\log\frac{x^2-x+1}{x^2+x+1} + C$$

10. $\int \frac{1}{x^4-1}\,dx$

Solution:

$= \int \frac{1}{(x^2)^2-1^2}\,dx$

$= \int \frac{1\cdot dx}{(x^2+1)(x^2-1)}$

Let $\frac{1}{(x^2+1)(x^2-1)} = \frac{Ax+B}{(x^2+1)} + \frac{Cx+D}{x^2-1}$

or, $1 = (Ax+B)(x^2-1) + (Cx+D)(x^2+1)$

or, $1 = Ax^3 - Ax + Bx^2 - B + Cx^3 + Cx + Dx^2 + D$

or, $1 = Ax^3 + Cx^3 + Bx^2 + Dx^2 + Cx - Ax + D - B$

or, $1 = (A+C)x^3 + (B+D)x^2 + (C-A)x + (D-B)$

On comparing like terms, we get:

$A + C = 0$ — (I)

$B + D = 0$ — (II)

$C - A = 0$ — (III)

$D - B = 1$ — (IV)

On solving all the above ~~eq~~ equations, we get

$A = 0$, $B = -\frac{1}{2}$, $C = 0$ and $D = \frac{1}{2}$

Then,

$\frac{1}{(x^2+1)(x^2-1)} = \frac{-1}{2(x^2+1)} + \frac{1}{2(x^2-1)}$

Integrating both sides;

$\int \frac{1\cdot dx}{(x^2+1)(x^2-1)} = \frac{1}{2}\left[\int \frac{1\cdot dx}{(x^2-1)} - \int \frac{1\cdot dx}{x^2+1}\right]$

$= \frac{1}{2}\left[\frac{\text{Log}(x-1)}{2.1(x+1)} - \text{Tan}^{-1}x\right] + C$

$= \frac{1}{4}\text{Log}\left(\frac{x-1}{x+1}\right) - \frac{\text{Tan}^{-1}x}{2} + C$

11. $\int \frac{x^2 \cdot dx}{x^4 - 2x^2 - 15}$

$= \int \frac{x^2}{(x^2+3)(x^2-5)}$

Let $\frac{x^2}{(x^2+3)(x-5)} = \frac{Ax+B}{(x^2+3)} + \frac{C}{(x-5)}$

or, $x^2 = (Ax+B)(x-5) + C(x^2+3)$

or, $x^2 = Ax^2 - 5Ax + Bx - 5B + Cx^2 + 3C$

or, $x^2 = (A+C)x^2 + (B-5A)x + (3C-5B)$

On Comparing the like terms, we get,

$A + C = 1$ — (i)

$B - 5A = 0$ — (ii)

$3C - 5B = 0$ — (iii)

On Solving them we get;

$A =$

Let $\frac{x^2}{(x^2+3)(x^2-5)} = \frac{Ax+B}{(x^2+3)} + \frac{Cx+D}{(x^2-5)}$

or $x^2 = (Ax+B)(x^2-5) + (Cx+D)(x^2+3)$

or, $x^2 = Ax^3 - 5Ax + Bx^2 - 5B + Cx^3 + 3Cx + Dx^2 + 3D$

or $x^2 = Ax^3 + Cx^3 + Bx^2 + Dx^2 + 3Cx - 5Ax + 3D - 5B$

or, $x^2 = (A+C)x^3 + (B+D)x^2 + (3C-5A)x + (3D-5B)$

On Comparing the like terms, we get;

$B + D = 1$ — (i)

$A + C = 0$ — (ii)

$3C - 5A = 0$ — (iii)

$3D - 5B = 0$ — (iv)

On Solving them we get;

$A = 0$, $B = \frac{3}{8}$, $C = 0$ and $D = \frac{5}{8}$

Now, $\frac{x^2}{(x^2+3)(x^2-5)} = \frac{3}{8(x^2+3)} + \frac{5}{8(x^2-5)}$

Integrating both sides;

$$\int \frac{x^2 \cdot dx}{(x^2+3)(x^2-5)} = \frac{3}{8}\int \frac{dx}{(x^2+3)} + \frac{5}{8}\int \frac{dx}{x^2-5}$$

$$= \frac{3}{8}\int \frac{dx}{x^2+(\sqrt{3})^2} + \frac{5}{8}\int \frac{dx}{x^2-(\sqrt{5})^2}$$

$$= \frac{3}{8}\cdot\frac{1}{\sqrt{3}}\cdot \text{Tan}^{-1}\left(\frac{x}{\sqrt{3}}\right) + \frac{5}{8}\cdot\frac{1}{2\cdot\sqrt{5}} \text{Log}\frac{x-\sqrt{5}}{x+\sqrt{5}} + C$$

$$= \frac{\sqrt{3}}{8}\text{Tan}^{-1}\frac{x}{\sqrt{3}} + \frac{\sqrt{5}}{16}\text{Log}\left(\frac{x-\sqrt{5}}{x+\sqrt{5}}\right) + C$$

12. $\int \frac{x^3 \cdot dx}{2x^4 - 3x^2 - 5}$

Solution:

$$\int \frac{x^3 \cdot dx}{(2x^2-5)(x^2+1)}$$

Let, $\frac{x^3}{(2x^2-5)(x^2+1)} = \frac{Ax+B}{(2x^2-5)} + \frac{Cx+D}{(x^2+1)}$

or, $x^3 = (Ax+B)(x^2+1) + (Cx+D)(2x^2-5)$

or, $x^3 = Ax^3 + Ax + Bx^2 + B + 2Cx^3 - 5Cx + 2Dx^2 - 5$

or, $x^3 = Ax^3 + 2Cx^3 + Bx^2 + 2Dx^2 + Ax - 5Cx + B - 5D$

or, $x^3 = (A+2C)x^3 + (B+2D)x^2 + (A-5C)x + (B-5D)$

On comparing them we get;

$A + 2C = 1$ —— (i)

$B + 2D = 0$ —— (ii)

$A - 5C = 0$ —— (iii)

$B - 5D = 0$ —— (IV)

On solving them, we get;

$A = \frac{5}{7}$, $B = 0$, $C = \frac{1}{7}$ and $D = 0$

So,

$$\frac{x^3}{(2x^2-5)(x^2+1)} = \frac{5x}{7(2x^2-5)} + \frac{1x}{7(x^2+1)}$$

Integrating both sides;

$$\int \frac{x^3}{(2x^2-5)(x^2+1)} = \frac{5}{7}\int \frac{x.dx}{2x^2-5} + \frac{1}{7}\int \frac{x.dx}{(x^2+1)}$$

$$= \frac{5}{7\times 4}\int \frac{4x.dx}{2x^2-5} + \frac{1}{7\times 2}\int \frac{2x.dx}{(x^2+1)}$$

$$= \frac{5}{28}\text{Log}(2x^2-5) + \frac{1}{14}\text{Log}(x^2+1) + C$$

13. $\int \frac{x^2\,dx}{(x^2+a^2)(x^2+b^2)}$

Solution:

Let $\frac{x^2}{(x^2+a^2)(x^2+b^2)} = \frac{Ax+B}{(x^2+a^2)} + \frac{Cx+D}{(x^2+b^2)}$

or, $x^2 = (Ax+B)(x^2+b^2) + (Cx+D)(x^2+a^2)$

or, $x^2 = Ax^3 + Ab^2x + Bx^2 + Bb^2 + Cx^3 + Ca^2x + Dx^2 + Da^2$

or, $x^2 = Ax^3 + Cx^3 + Bx^2 + Dx^2 + Ab^2x + Ca^2x + Bb^2 + Da^2$

or, $x^2 = (A+C)x^3 + (B+D)x^2 + (Ab^2+Ca^2)x + Bb^2 + Da^2$

On Comparing the like terms we get

$A + C = 0$ —— (I)

$B + D = 1$ —— (II)

Date: / /

$Ab^2 + Ca^2 = 0$ —— (III)

$Bb^2 + Da^2 = 0$ —— (IV)

On solving them we get;

$$A = 0, \quad B = \frac{a^2}{a^2-b^2}, \quad C = 0, \quad D = \frac{-b^2}{a^2-b^2}$$

Then,

$$\frac{x^2}{(x^2+a^2)(x^2+b^2)} = \frac{a^2}{(a^2-b^2)(x^2+a^2)} - \frac{b^2}{(a^2-b^2)(x^2+b^2)}$$

Integrating both sides;

$$\int \frac{x^2 \cdot dx}{(x^2+a^2)(x^2+b^2)} = \frac{a^2}{a^2-b^2}\int \frac{dx}{x^2+a^2} - \frac{b^2}{a^2-b^2}\int \frac{dx}{x^2+b^2}$$

$$= \frac{a^2}{(a^2-b^2)} \cdot \frac{1}{a} \tan^{-1}\frac{x}{a} - \frac{b^2}{(a^2-b^2)} \cdot \frac{1}{b} \tan^{-1}\frac{x}{b}$$

$$= \frac{a}{a^2-b^2} \tan^{-1}\frac{x}{a} - \frac{b}{a^2-b^2} \tan^{-1}\frac{x}{b} + C$$

14. $\int \frac{x^3 \cdot dx}{(x^2+a^2)(x^2+b^2)}$

Solution:

Let $\frac{x^3}{(x^2+a^2)(x^2+b^2)} = \frac{Ax+B}{(x^2+a^2)} + \frac{Cx+D}{(x^2+b^2)}$

or, $x^3 = (Ax+B)(x^2+b^2) + (Cx+D)(x^2+a^2)$

or, $x^3 = Ax^3 + Ab^2x + Bx^2 + Bb^2 + Cx^3 + Cxa^2 + Dx^2 + Da^2$

or, $x^3 = Ax^3 + Cx^3 + Bx^2 + Dx^2 + Ab^2x + Cxa^2 + Bb^2 + Da^2$

or, $x^3 = (A+C)x^3 + (B+D)x^2 + (Ab^2 + Ca^2)x + (Bb^2 + Da^2)$

On Comparing the common terms, we get;

$A + C = 1$ — (I)

$B + D = 0$ — (II)

$Ab^2 + Ca^2 = 0$ — (III)

$Bb^2 + Da^2 = 0$ — (IV)

Solving them we get;

$A = \frac{a^2}{a^2 - b^2}$, $B = 0$, $C = \frac{-b^2}{a^2 - b^2}$

$D = 0$

Then,

$$\frac{x^3}{(x^2+a^2)(x^2+b^2)} = \frac{a^2 x}{(a^2-b^2)(x^2+a^2)} - \frac{b^2 x}{(a^2-b^2)(x^2+b^2)}$$

Integrating both sides;

$$\int \frac{x^3}{(x^2+a^2)(x^2+b^2)} = \frac{a^2}{a^2-b^2}\int \frac{x\,dx}{x^2+a^2} - \frac{b^2}{(a^2-b^2)}\int \frac{x.dx}{x^2+b^2}$$

$$= \frac{a^2}{2(a^2-b^2)}\int \frac{2x.dx}{x^2+a^2} - \frac{b^2}{2(a^2-b^2)}\int \frac{2x.dx}{x^2+b^2}$$

$$= \frac{1}{2(a^2-b^2)}\left[a^2 \log(x^2+a^2) - b^2 \log(x^2+b^2)\right]$$

15. $\int \frac{dx}{(x-1)^2(x-2)^3}$

Solution:

Put $x - 1 = z(x-2)$ — (I)

or, $x - 1 = xz - 2z$

or, $2z - 1 = xz - x$

or, $2z - 1 = x(z-1)$

or, $x = \frac{2z-1}{(z-1)}$

$$\text{or, } \frac{dx}{dz} = \frac{(z-1)(2) - (2z-1)}{(z-1)^2}$$

$$\text{or, } \frac{dx}{dz} = \frac{2z - 2 - 2z + 1}{(z-1)^2}$$

$$\text{or, } dx = \frac{-1}{(z-1)^2} dz$$

Also,

$$\int \frac{dx}{(x-1)^2(x-2)^2} = \int \frac{-dz}{(z-1)^2 \left(\frac{2z-1}{z-1} - 1\right)^2 \left(\frac{2z-1}{z-1} - 2\right)^3}$$

$$= \int \frac{-dz}{(z-1)^2 \left(\frac{2z-1-z+1}{z-1}\right)^2 \left(\frac{2z-1-2z+2}{z-1}\right)^3}$$

$$= \int \frac{-dz}{(z-1)^2 \left(\frac{z}{z-1}\right)^2 \left(\frac{1}{z-1}\right)^3}$$

$$= \int \frac{-(z-1)^3 \cdot dz}{z^2}$$

$$= \int \frac{-(z^3 - 3z^2 + 3z - 1)\, dz}{z^2}$$

$$= \int \frac{(-z^3 + 3z^2 - 3z + 1)\, dz}{z^2}$$

$$= \int \left(-z + 3 - \frac{3}{z} + z^{-2}\right) dz$$

$$= \frac{-z^2}{2} + 3z - 3\log z - \frac{1}{z} + c$$

Now,

Using the value of 'z' from equation (1)

$$= -\frac{1}{2}\left(\frac{x-1}{x-2}\right)^2 + 3\left(\frac{x-1}{x-2}\right) - 3\log\left(\frac{x-1}{x-2}\right) - \left(\frac{x-2}{x-1}\right) + c$$